Configuration Space and Degrees of Freedom

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Where this fits:

- **Builds on:** Pose; Why exact planning is hard.
- **Leads to:** Obstacle region and free space; Complete, resolution-complete and probabilistically complete; Manipulation planning (transit and transfer); Kinodynamic planning and phase space.

1. Overview

Key point: The routes of the earlier Notes move a point from cell to cell, but the robot is a body with a size and a heading, and placing it takes three numbers, (x, y, θ). The numbers that place every point of a robot are its configuration; all configurations together form the configuration space, and planning a motion means finding a path through that space.

The robot can now plan a task. The task planner of the cup task returns four actions, and each move is a route on the 0.5 m grid, such as the shortest route from the start cell to the cell just inside the door, 3.414 m long (routes of the plan).

What still breaks is the robot itself (Figure 1). The route moves a point from cell centre to cell centre. The real robot is a body: seen from above, a disk of radius 0.25 m centred on the middle of its axle, big enough to hold the 0.2 m axle and the wheels. At the start it stands at (2 m, 1 m) and faces 30°. Two problems follow:

- the two numbers (x, y) say where the centre is, but not which way the robot faces, and a robot facing the table drives off differently from one facing the door;
- the heading wraps round: 350° and 10° are only 20° apart, so a planner that treats the heading as an ordinary number thinks the two are 340° apart.

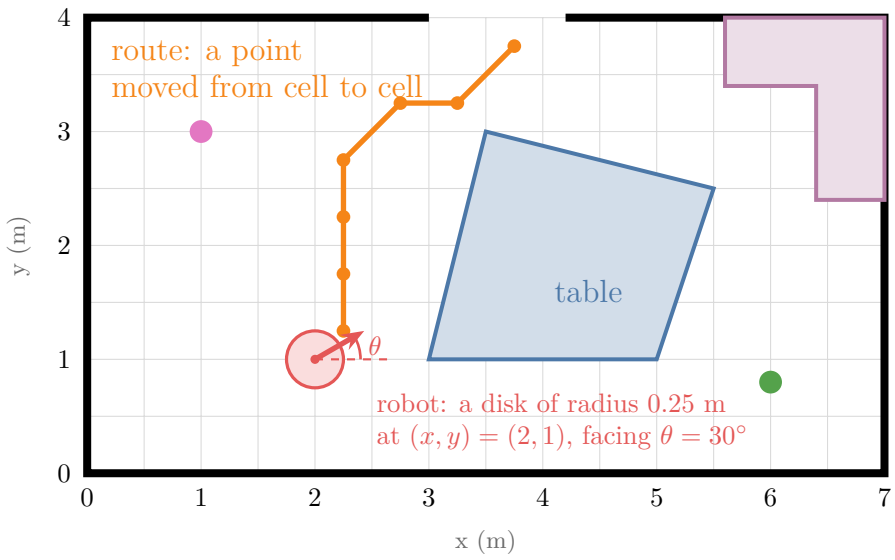


Figure 1: The room with the table, the cabinet and the two poles. Orange: a route of the graph-search Notes, a point moved from cell centre to cell centre towards the door. Red: the robot, a disk of radius 0.25 m centred at (2, 1), with an arrow for its heading of 30 degrees measured from the x-axis

In this Note we answer three questions:

1. Which numbers place the whole robot, and how many are needed (Section 2 and Section 3)?
2. What shape do all those numbers form, and how do we write them down without a jump at 360° (Section 4 and Section 5)?
3. What is the planning problem in that space (Section 6)?

A second robot joins the floor robot in this Note: an arm with two joints on the robot's top plate, the kind of arm that picks up the cup. Each of its two joints turns all the way round, so it shows a shape that the floor robot's single angle cannot (the 2R arm of MR §2.3.1, R for a revolute joint). Two facts from earlier work are used throughout. A robot arm is a chain of rigid links joined by joints, and a revolute joint lets one link turn about the other. And once the position and heading of a rigid body are known, the position of each of its points follows by a rotation by the heading and a shift by the position.

2. Configuration and configuration space

> **Key point:** The first question is which numbers place the whole robot. For the floor robot, the centre (x, y) and the heading θ fix every point of its body, so (2, 1, 0.524) is its configuration; all configurations together form the configuration space, a box whose top face and bottom face are the same headings.

2.1 The configuration: numbers that place every point

The route gives two numbers, the (x, y) of a point. The robot's body is rigid: every point of it keeps its distance from every other point. So once we know where the centre is and which way the robot faces, every other point of the body is fixed: rotate the point by the heading, then shift it by the centre's position.

Take the front point of the body, 0.25 m ahead of the centre:

- x, y are the centre's position, 2 m and 1 m;
- θ is the heading, 30°, which is 0.524 rad;
- a is how far the point lies ahead of the centre, in the robot's own frame, 0.25 m;
- x_f, y_f are the front point's position in the room, in m.

The point lies a along the heading, and a length along the heading splits into a run along x and a rise along y, the same split as the robot's speed in cosine and sine parts:

$$x_f = x + a\cos\theta$$

$$= 2 + 0.25 \times 0.8660$$

$$= 2 + 0.2165 = 2.2165 \text{ m}$$

$$y_f = y + a\sin\theta$$

$$= 1 + 0.25 \times 0.5 = 1.125 \text{ m}$$

Any other point of the body works the same way, so three numbers fix all of them. A list of numbers that fixes the position of every point of a robot is its **configuration** (G-2286) (MR Definition 2.1). We write it as one **vector** (G-2081), a list of numbers, named q:

$$q = (x,\ y,\ \theta)$$

$$q = (2,\ 1,\ 0.524)$$

For the floor robot the configuration is its **pose** (G-2280), the position and heading of RO-001, because the body is rigid. For the arm, the two joint angles fix every point: with joint 1 turned to $\theta_1 = 30°$ and joint 2 to $\theta_2 = 60°$, its configuration is

$$q = (\theta_1,\ \theta_2) = (30°,\ 60°)$$

and, with links 0.3 m and 0.2 m long, the hand sits at (0.260 m, 0.350 m) from the shoulder (notebook; the arm is drawn in Section 3.2). One configuration is one placement of the robot; a planner needs all of them at once.

2.2 The configuration space of the floor robot

A planner has to consider every placement of the robot, not just the present one. All configurations of a robot together form its **configuration space** (G-2287), or C-space for short (MR Definition 2.1). Each point of the C-space is one placement of the whole robot.

Figure 2 draws the floor robot's C-space as a box: the room's floor across, x from 0 to 7 m and y from 0 to 4 m, and the heading θ upward, from 0 to 2π. The robot at (2, 1, 0.524) is one point of the box. The same robot turned round on the spot, facing 180° (3.142 rad), is the point straight above it: a different configuration, although the centre has not moved.

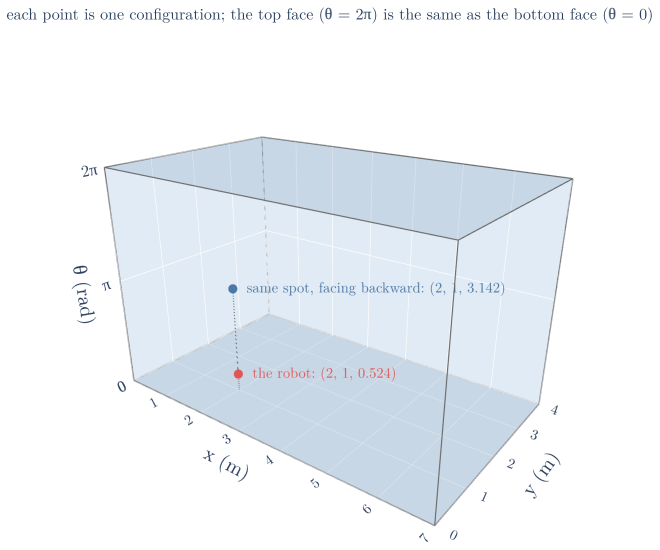


Figure 2: The configuration space of the floor robot as a box: x from 0 to 7 m and y from 0 to 4 m across, the heading theta from 0 to 2 pi upward. The robot at (2, 1, 0.524) is a red point near the floor of the box; the same spot facing the other way, (2, 1, 3.142), is a blue point straight above it. The top face (theta = 2 pi) and the bottom face (theta = 0) are shaded: they are the same headings

The box has one feature a room does not. Turning by a full 2π brings the robot back to the same heading, so the top face, $\theta = 2\pi$, holds the same configurations as the bottom face, $\theta = 0$. A point that climbs out through the top comes back in through the bottom. The glued faces are the wrap-around of headings, the **angle wrapping** (G-2404) of RO-012, seen in the C-space.

Figure 3 plays it. The robot drives once round the circle of radius 0.5 m of RO-001, at 0.5 m/s while turning at 1 rad/s. Watch the right panel: the point climbs as the heading turns,

leaves through the top face in the frame "t = 5.74 s", is back at the bottom face in the frame "θ passed 2π", and in the last frame stands where it started. The time at which the heading reaches 2π:

- θ_0 is the start heading, 0.5236 rad;
- ω is the angular velocity, 1 rad/s;
- t is the time, in s.

Our numbers first, each followed by the same step in symbols:

$$0.5236 + 1 \times t = 6.2832$$

$$\theta_0 + \omega\, t = 2\pi$$

$$t = (6.2832 - 0.5236)/1 = 5.7596 \text{ s}$$

$$t = (2\pi - \theta_0)/\omega$$

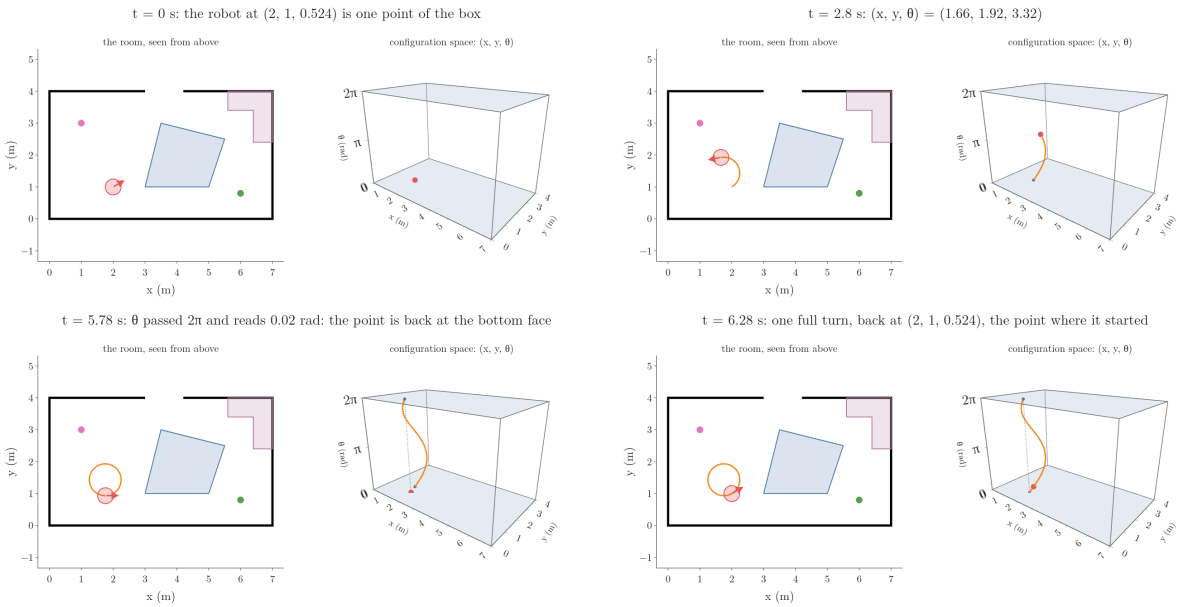


Figure 3: The robot drives RO-001's circle of radius 0.5 m for one full turn, 6.28 s. Left: the room seen from above, the robot as a red disk of radius 0.25 m with its heading arrow and its orange trail. Right: the configuration space box; the red point climbs a curve as the heading grows. In the frame 't = 5.74 s' the point is about to leave the top face; in the frame 'θ passed 2π' it is back at the bottom face, and a grey dashed line joins the exit and the entry, which are the same configuration. Last frame: after one full turn the robot is back at (2, 1, 0.524), the point where it started

So the C-space of the floor robot is a box with its top and bottom glued together. The box needs three numbers per point. Section 3 asks whether three is the right count, and how to count for any robot.

3. Degrees of freedom

> **Key point:** How many numbers are needed, the rest of the first question, is a robot's degrees of freedom, the smallest number of numbers that places it: 3 for the floor robot and 2 for the arm. We count them as freedoms minus independent constraints, point by point for one body (2, 1 and 0 freedoms for three of its points), and with Grübler's formula for links joined by joints.

3.1 Counting point by point

Section 2 placed the floor robot with three numbers. A body has endless points, each with two numbers of its own in the plane, so why exactly three? We count point by point (MR §2.1), on three points of the robot's body:

- A, the centre, at (2, 1);
- B, 0.25 m ahead of A, at (2.2165, 1.125), the front point of Section 2.1;
- C, 0.25 m to the left of A, at (1.875, 1.217), and 0.354 m from B.

Figure 4 places them one at a time. Watch the table on the right fill in, one row per point:

- frame "A, the centre": A can be anywhere in the plane: 2 numbers, no rule, so 2 freedoms;
- frame "B stays 0.25 m from A": B has 2 numbers, but one rule, its fixed distance to A, keeps it on the orange circle round A. Only its place on the circle is free, 1 freedom, and sliding B round the circle turns the body;
- frame "C stays 0.25 m from A and 0.354 m from B": two rules keep C on two circles, which meet in only 2 points, and one of them is the body flipped upside down, which a floor robot cannot do. So C is fixed: 0 freedoms.

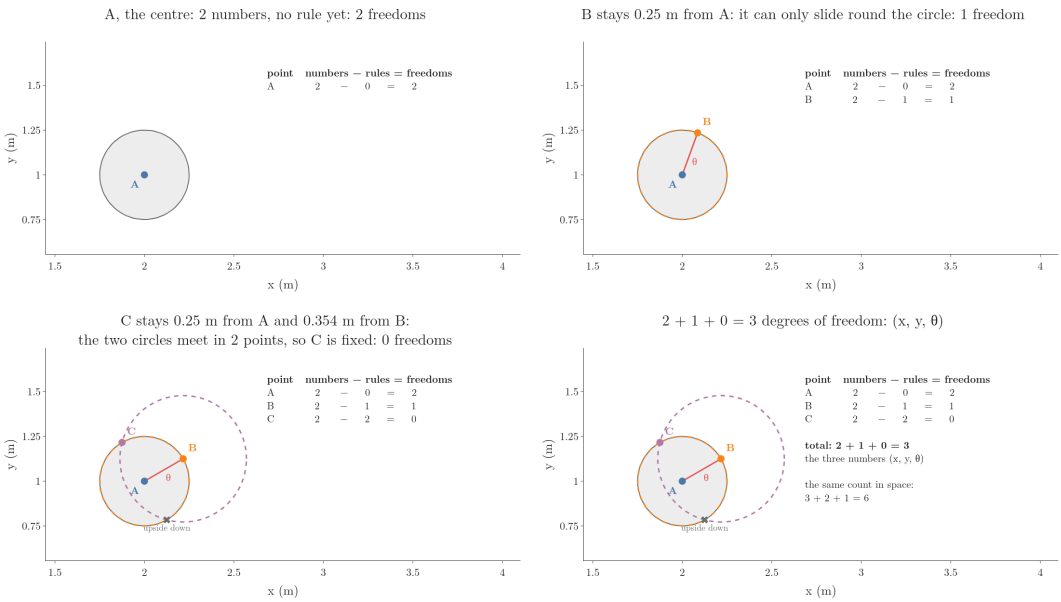


Figure 4: Close-up of the robot's body, a grey disk of radius 0.25 m at (2, 1), with a table of 'numbers − rules = freedoms'. Frame 'A, the centre': A (blue) at the centre, row A: 2 − 0 = 2. Frames 'B stays 0.25 m from A': B (orange) slides round the orange dashed circle of radius 0.25 m about A, and the red line from A to B turns with it, its direction being the heading θ; row B: 2 − 1 = 1. Frame 'C stays 0.25 m from A and 0.354 m from B': the purple dashed circle about B meets the circle about A at C and at a grey cross marked 'upside down'; row C: 2 − 2 = 0. Last frame: total 2 + 1 + 0 = 3, the three numbers (x, y, θ); the same count in space, 3 + 2 + 1 = 6

Any further point is fixed by its distances to A, B and C, so it adds nothing. The total:

$$2 + 1 + 0 = 3$$

A's two numbers are x and y, and B's place on its circle is the heading θ. The smallest number of real numbers needed to place a robot is its **degrees of freedom** (G-2310), DOF for short; it is also the number of dimensions of its C-space, 3 for the box of Figure 2 (MR Definition 2.1).

The general rule (MR eq. 2.1):

- F is the sum of the freedoms of the points, before any rule;
- c is the number of independent constraints, rules that fix how the points may lie relative to each other;
- DOF is the degrees of freedom.

$$\text{DOF} = F - c$$

For the three points of the body in the plane, and then the same body in space, where each point has 3 numbers:

$$F = 3 \times 2 = 6$$

$$\text{DOF} = 6 - 3 = 3$$

$$F = 3 \times 3 = 9$$

$$\text{DOF} = 9 - 3 = 6$$

Point by point, as in the last frame of Figure 4, the body in space gives the same 6: A has 3 freedoms, B 2 and C 1.

$$3 + 2 + 1 = 6$$

A body that moves in space, such as a flying drone, has 6 degrees of freedom: 3 numbers for its position and 3 angles (MR §2.1).

"Independent" matters. A fourth point D of the body in the plane has 3 distance rules, to A, B and C, but once two of them hold the third holds too, so only 2 are independent, and D's 2 numbers minus its 2 independent rules leave it no freedom (MR §2.1). A rule that follows from the others removes nothing.

The wheels' rule of RO-001, that the robot cannot slide sideways, is not a constraint of this kind. The rule limits which way the robot can move at each moment, not where it can be, so the floor robot keeps all 3 degrees of freedom (MR §2.4). Counting point by point works for one body; a robot of several links needs a quicker count.

3.2 Degrees of freedom of a robot made of links

The arm of the Overview is our example of a robot made of several links: link 1 turns on the robot's top plate at joint 1, and link 2 turns on link 1 at joint 2 (Figure 5, left). Both are revolute joints.

- Each link moving freely in the plane would have 3 freedoms, like the robot's body in Section 3.1.
- A revolute joint lets one link only turn about the other: it leaves 1 of the 3 freedoms and so removes 2 (MR §2.2.1).

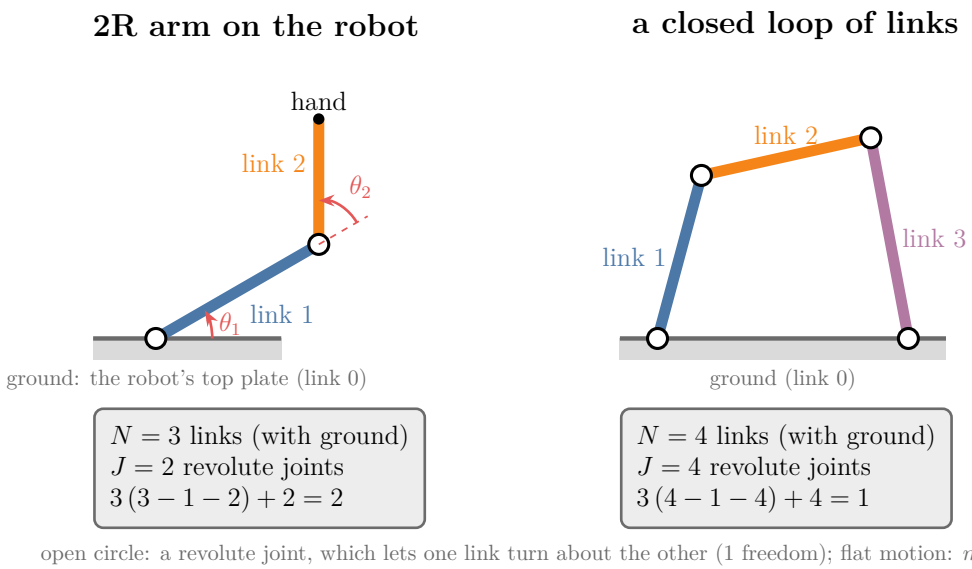


Figure 5: Left: the 2R arm on the robot. The robot's top plate is the ground (link 0); link 1 (blue) turns about joint 1 at angle theta1, link 2 (orange) turns about joint 2 at angle theta2, ending at the hand; joints are open circles. N = 3 links with the ground, J = 2 revolute joints: 3 (3 − 1 − 2) + 2 = 2. Right: a closed loop, three links (blue, orange, purple) and the ground joined end to end by 4 revolute joints. N = 4, J = 4: 3 (4 − 1 − 4) + 4 = 1

For the arm, the count goes:

- m is the number of freedoms of one body, 3 in the plane;
- N is the number of links, counting the ground (here the top plate) as a link: 3; the ground does not move, so only the other 2 links have freedoms to lose;
- J is the number of joints, 2;
- f_i is the number of freedoms joint i leaves, 1 for a revolute joint;
- $\sum$ means add up, here over the joints i from 1 to J.

Our numbers first, each followed by the same step in symbols:

$$\text{DOF} = 3 \times 2 - (3-1) - (3-1)$$

$$\text{DOF} = m\,(N-1) - \sum_{i=1}^{J} (m - f_i)$$

$$= 3 \times 2 - 3 \times 2 + 1 + 1$$

$$= m\,(N-1) - m\,J + \sum_{i=1}^{J} f_i$$

$$= 3\,(3 - 1 - 2) + 2 = 2$$

$$= m\,(N - 1 - J) + \sum_{i=1}^{J} f_i$$

The last line is **Grübler's formula** (G-2620) (MR eq. 2.4). The arm has 2 degrees of freedom, its two joint angles (θ_1, θ_2).

Two checks:

- The floor robot is one body and no joint, $N = 2$ (the floor and the body) and $J = 0$: the formula gives 3 (first line below), as in Section 3.1.
- The closed loop of Figure 5, right, has 3 moving links joined end to end, with both ends pinned to the ground, so 4 joints, and the formula gives 1 (second line below). Pinning the end of the third link to the ground removes 2 more freedoms (MR §2.2.2).

$$3\,(2 - 1 - 0) + 0 = 3$$

$$3\,(4 - 1 - 4) + 4 = 1$$

The formula holds only when the joints' constraints are independent; when some of them follow from the others, the true count is higher than the formula says (MR §2.2.2). The robot carrying the arm has the 3 degrees of freedom of its body plus the 2 of the arm, 5 in all.

The counts say how many numbers place each robot: 3 for the floor robot, 2 for the arm. They do not say what shape those numbers form, and both of the arm's angles wrap round, which Section 4 takes up.

4. The shape of a configuration space

> **Key point:** The second question is what shape all those numbers form: a heading lives on a circle, not on a line, and the arm's two angles live on a torus, the surface of a doughnut. Two spaces have the same shape when one can be bent or stretched into the other without cutting or gluing.

4.1 Same shape: bending allowed, cutting and gluing not

Section 3 counted the numbers, but the count is not the whole story. The heading is one number, like the position of a cart on a rail. Yet on a rail, 3.5 m and 0.1 m are 3.4 m apart, while the headings 350° and 10° are 20° apart. The difference is the shape: a heading's values close up into a circle, while a rail's values run along a line with nothing joining the ends.

Figure 6 makes "same shape" precise. Watch the circle of headings in the frames "stretch and bend the circle": it becomes an ellipse and a squashed loop, but it stays one closed loop,

so it is still the same shape. In the frames "cut at $\theta = 0$ and open it out", the circle has to be cut before it can be laid straight, and the cut makes two ends that the circle did not have.

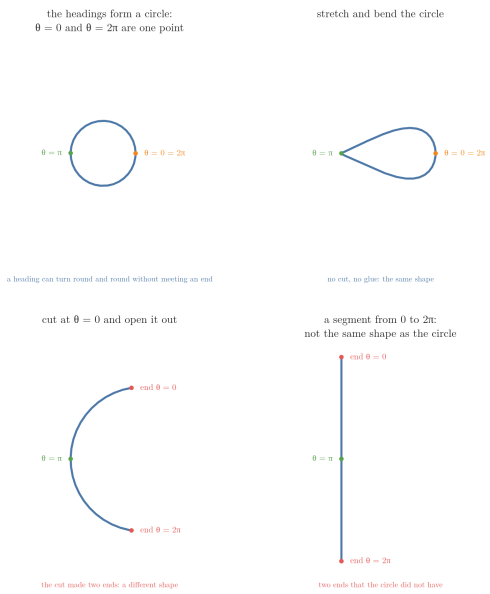


Figure 6: The circle of headings, with $\theta = \pi$ marked green and $\theta = 0 = 2\pi$ marked orange. Frames 'stretch and bend the circle': it becomes an ellipse and then a squashed loop, still closed: no cut, no glue, the same shape. Frames 'cut at $\theta = 0$ and open it out': the circle opens at $\theta = 0$ into an arc and then a straight segment, with red ends marked $\theta = 0$ and $\theta = 2\pi$. Last frame: a segment from 0 to 2π, not the same shape as the circle: two ends the circle did not have

Two spaces are **topologically equivalent** (G-2622) when one can be deformed into the other by bending and stretching, without cutting or gluing (MR §2.3.1). A coffee mug and the surface of a doughnut are equivalent this way: the handle's hole is the doughnut's hole. Three shapes of one dimension appear in robots (MR §2.3.1):

- the circle, written S^1: a heading, or a revolute joint that turns all the way round;
- the line, written $\mathbb{R}$, all real numbers: a sliding joint with no end stops;
- the closed interval, written $[a, b]$, the numbers from a to b with both ends included: a sliding joint with end stops, such as one that slides from 0 to 0.5 m.

None of the three is equivalent to another: the circle has no end, the line has no end but does not close up, and the interval has two ends. Our robots have more than one number, so their spaces are built from these shapes.

4.2 Building bigger spaces: the Cartesian product

The floor robot has three numbers: x and y each run along a line, and θ runs round a circle. What shape do the three make together? Every choice of (x, y) can be combined with every heading, so the space holds every pair of a point of the plane with a point of the circle.

Taking every pair of one point from each of two spaces makes a new space, their **Cartesian product** (G-2625), written with $\times$; its number of dimensions is the sum of theirs (MR §2.3.1; LaValle §4.1.2). In symbols:

- $\mathbb{R}^2$ is the plane, all pairs of real numbers such as (2, 1), itself the product $\mathbb{R} \times \mathbb{R}$;
- S^1 is the circle of headings;
- $\mathbb{R}^2 \times S^1$ is every pair of a point (x, y) with a heading θ, such as (2, 1, 0.524).

The C-spaces of our robots and of two others (MR §2.3.1):

Robot	C-space	Dimensions
floor robot	$\mathbb{R}^2 \times S^1$: the box of Figure 2 with top and bottom glued	3
the 2R arm	$S^1 \times S^1$: a torus	2
a knob that slides and turns	$\mathbb{R} \times S^1$: a cylinder	2
the floor robot carrying the arm	$\mathbb{R}^2 \times S^1 \times S^1 \times S^1$	5

The arm's product needs a picture. Figure 7 builds it: watch joint 2's angle drawn as a small circle, then joint 1's angle as a big circle. At every angle of joint 1 there is a whole circle of angles of joint 2, so copies of the small circle are placed round the big one, 30° apart in the frames "copies 30° apart", until in the frame "the copies sweep out the surface of a torus" they close into a doughnut surface. In the last frame, one point of the surface, the red one, is the arm at (30°, 60°).

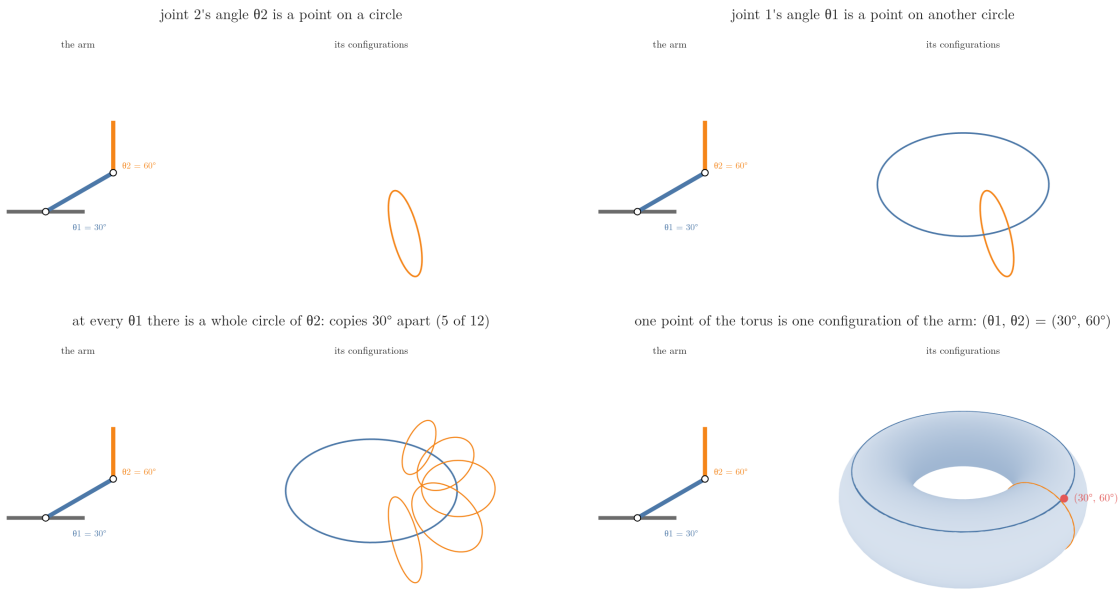


Figure 7: Left: the arm at $\theta 1 = 30°$, $\theta 2 = 60°$. Right, frame by frame: joint 2's angle as an orange circle; joint 1's angle as a blue circle; orange copies of the small circle placed round the blue one, 30 degrees apart, 1 to 12 of them; then the shaded surface they sweep out, a torus. Last frame: on the torus, the orange circle through $\theta 1 = 30°$, the blue circle through $\theta 2 = 60°$, and where they cross, the red point (30°, 60°)

The surface is the **torus** (G-2624), T^2, the product $S^1 \times S^1$: every point of it is one configuration of the arm, and every configuration of the arm is one point of it (MR §2.1). The product of two circles is a torus, not the surface of a ball (a sphere): a sphere and a torus are not topologically equivalent (MR §2.3.1). The spaces are now built; what they look like close to one configuration is the next question.

4.3 Manifolds: every small patch looks flat

Section 4.2 built curved and wrapped spaces, but a robot moves in small steps. What does a C-space look like close to one configuration? Figure 8 zooms in on one point of three shapes. A small piece of the circle looks like a piece of a straight line, and a small patch of the torus looks like a piece of a flat plane. At the joint of a T, three pieces of line meet, which no single line looks like.

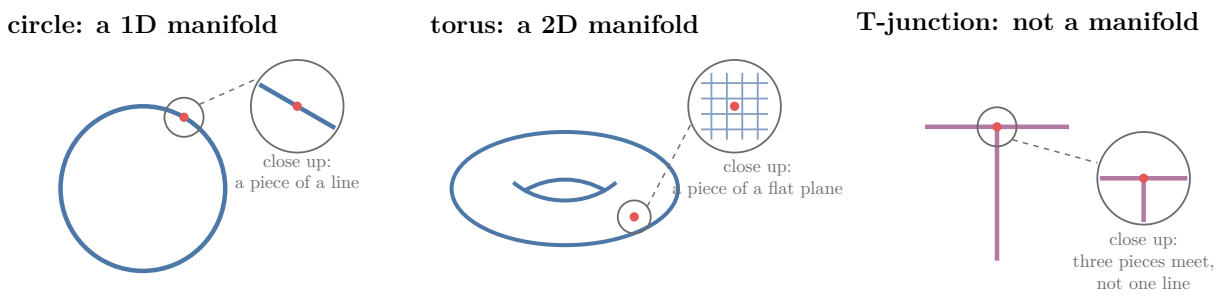


Figure 8: Three shapes with a zoomed close-up of one red point each. Circle: the close-up is a piece of a straight line, so the circle is a 1D manifold. Torus: the close-up is a piece of a flat grid, so the torus is a 2D manifold. T-junction: the close-up at the joint shows three pieces of line meeting, not one line, so the T is not a manifold

A space in which the neighbourhood of every point looks like a piece of flat space of the same number of dimensions, a line for one dimension, a plane for two, is a **manifold** (G-2626) (LaValle §4.1.2). Near any point of a manifold, "the set of directions that one can move appears the same" as in flat space (LaValle §4.1.2), so a robot at any configuration can move a little in each of its directions, 3 for the floor robot. The C-spaces of this Note are all manifolds: the circle, and the cylinder, the torus and the floor robot's $\mathbb{R}^2 \times S^1$, which are built as Cartesian products of manifolds (LaValle §4.1.2).

A manifold is flat only up close. Its overall shape, the circle closing up or the torus wrapping round in two directions, still matters as soon as we write a configuration down as numbers, which is the subject of Section 5.

5. Writing a configuration as numbers

> **Key point:** The rest of the second question is how to write a configuration down without a jump at 360°. The fewest numbers, θ for a heading and (θ1, θ2) for the arm, jump at 2π, because the circle and the torus must be cut open to lay them flat; more numbers with a rule, such as ($\cos\theta$, $\sin\theta$) with $\cos^2\theta + \sin^2\theta = 1$, never jump.

5.1 Cutting the torus into a square

The arm's configurations form a torus, but a program stores two numbers, θ_1 and θ_2, each from 0 to 2π. Those pairs fill a square. Figure 9 shows how the square and the torus are related. Watch the frame "roll it until the orange edges meet": the square rolls into a tube, because $\theta_2 = 0$ and $\theta_2 = 2\pi$ are the same angle of joint 2. In the frames "stretch and bend the tube", the tube bends round until its blue ends, $\theta_1 = 0$ and $\theta_1 = 2\pi$, meet. Run backwards, the film cuts the torus twice to lay it flat as the square (MR §2.3.1).

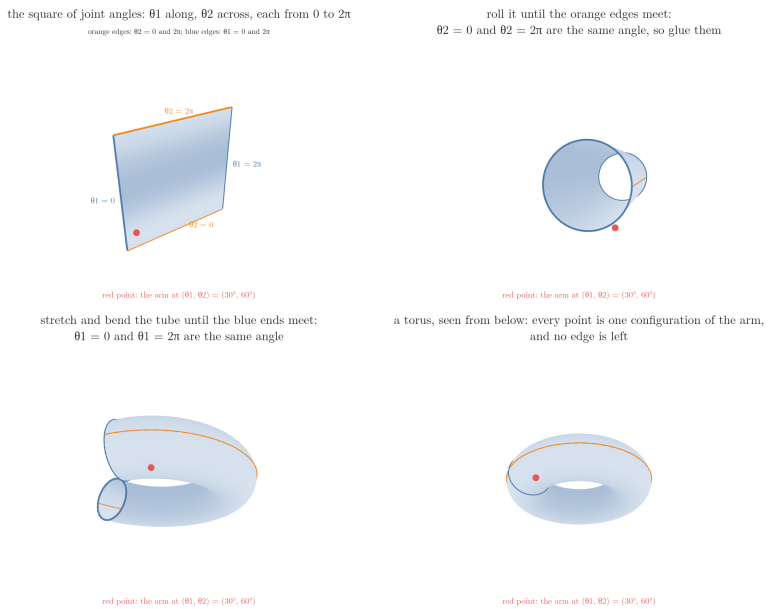


Figure 9: Frame 'the square of joint angles': a flat square, $\theta 1$ along and $\theta 2$ across, each from 0 to 2π; its edges $\theta 2 = 0$ and $\theta 2 = 2\pi$ are orange, its edges $\theta 1 = 0$ and $\theta 1 = 2\pi$ blue; the red point is the arm at (30°, 60°). Frames 'roll it until the orange edges meet': the square curls into a tube with the orange edges joined. Frames 'stretch and bend the tube until the blue ends meet': the tube bends into a ring. Last frame: a torus, seen from below, with no edge left, and the red point on it

The cut has a price. Figure 10 moves the arm smoothly, θ_1 from 300° on past 360° to 60°, while θ_2 goes from 60° to 120°. Watch the middle panel: in the frame "$\theta 1$ passes 360°", the point in the square jumps from the right edge to the left edge, while on the torus, right, it moves on without a jump. The last frame shows the square's path cut in two and the torus path in one piece.

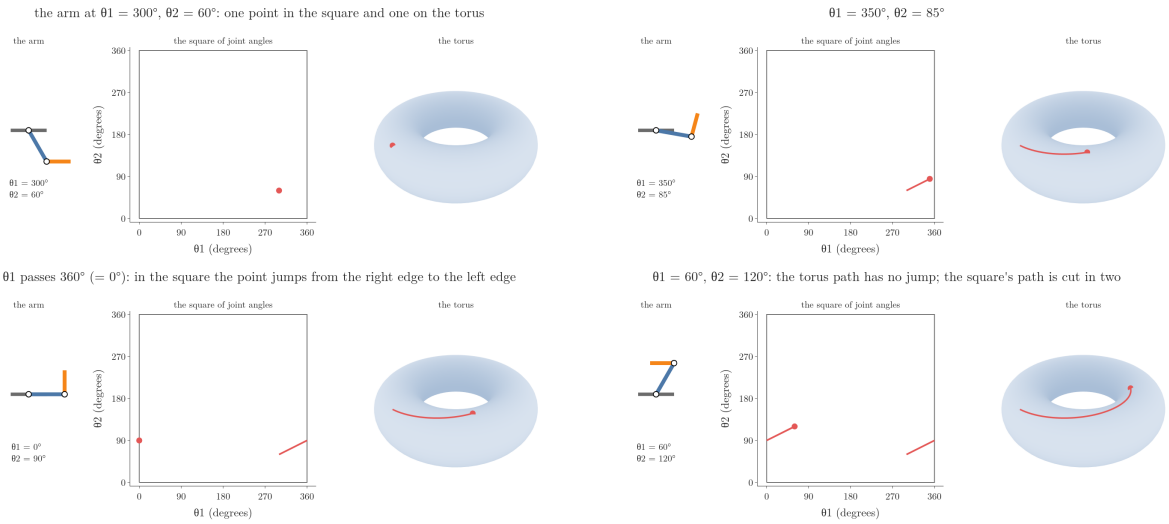


Figure 10: Left: the arm, with its joint angles written below it. Middle: the square of joint angles, $\theta 1$ across and $\theta 2$ up, each from 0 to 360 degrees, with the arm's point and its red trail. Right: the torus with the same point and trail. First frame: $\theta 1 = 300°$, $\theta 2 = 60°$. Frame '$\theta 1$ passes 360°': the square's point jumps from the right edge to the left edge; on the torus the trail is unbroken. Last frame: $\theta 1 = 60°$, $\theta 2 = 120°$; the square's path is in two pieces, the torus path in one

The jump belongs to the numbers, not to the arm: the arm moved smoothly, and the torus shows it (MR §2.3.1). Writing a space with exactly as many numbers as it has dimensions, such as (θ_1, θ_2) for the torus or latitude and longitude for the Earth's surface, is an **explicit parametrization** (G-2627) of it (MR §2.3.2). An explicit parametrization uses the fewest numbers, but for the torus and the sphere it breaks somewhere: a jump at 2π, or a pole (MR §2.3.1, §2.3.2). Section 5.2 removes the jump with more numbers.

5.2 More numbers, no jump: the implicit representation

The jump of Section 5.1 in numbers, for the heading, which turns 2° from 359° to 1°:

- θ is the explicit coordinate, in rad, from 0 to 2π.

$$\theta(359°) = 6.26573$$

$$\theta(1°) = 0.01745$$

$$0.01745 - 6.26573 = -6.24828 \text{ rad}$$

A turn of 0.0349 rad shows up as a change of -6.24828 rad. The fix is to write the heading as the point where its direction meets a circle of radius 1. Two numbers with one rule:

- $\cos\theta$ and $\sin\theta$ are the two numbers, the point's distance across and up from the circle's centre;
- the rule is that the point lies on the circle of radius 1.

$$\cos^2\theta + \sin^2\theta = 1$$

By the rule of Section 3.1, 2 numbers minus 1 independent rule leave 1 degree of freedom, as a heading should. At the same two headings (notebook):

$$(\cos 359^\circ, \sin 359^\circ) = (0.9998, -0.0175)$$

$$(\cos 1^\circ, \sin 1^\circ) = (0.9998, 0.0175)$$

The point moves 0.035, the size of the 2° turn, and nothing jumps. The same pair settles the second problem of the Overview, the headings 350° and 10°. As explicit numbers they lie far apart:

$$\theta(350^\circ) - \theta(10^\circ)$$

$$= 6.10865 - 0.17453 = 5.93412 \text{ rad}$$

As implicit pairs they lie close together:

$$(\cos 350^\circ, \sin 350^\circ) = (0.98481, -0.17365)$$

$$(\cos 10^\circ, \sin 10^\circ) = (0.98481, 0.17365)$$

$$0.17365 - (-0.17365) = 0.34730$$

The two points differ only in their second number, $\sin\theta$, by 0.3473, about the 0.349 rad of a 20° turn, so a planner that compares the pairs sees two nearby headings. Figure 11 turns the heading from 330° to 30°. Watch the two plots on the right: in the frame "heading 360° = 0°" the explicit θ drops from 2π to 0, while $\cos\theta$ and $\sin\theta$ move on smoothly; the last frame shows one jump in θ and none in the other two.

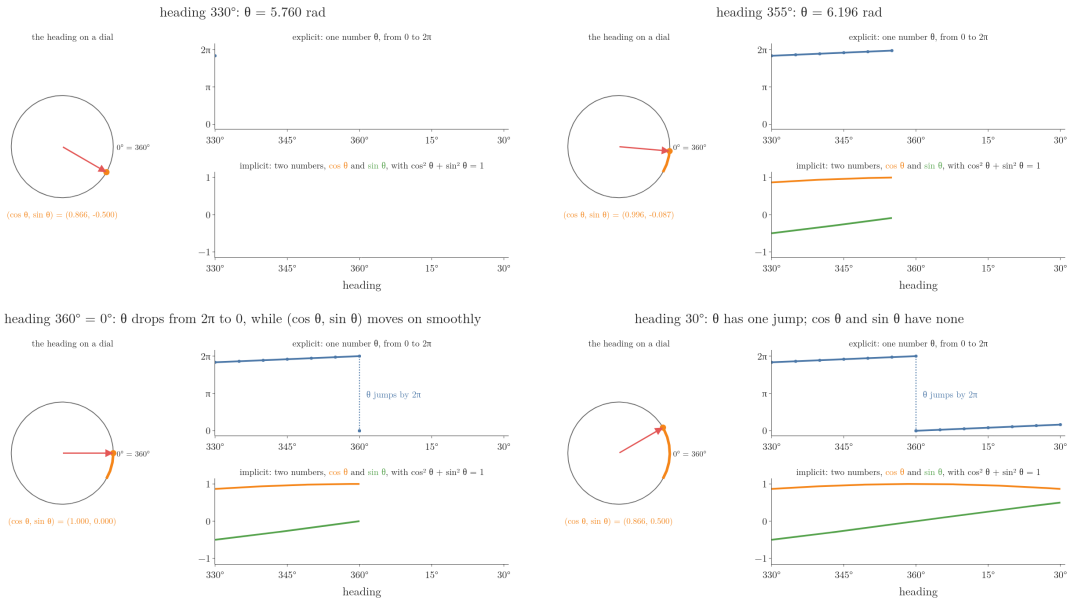


Figure 11: Left: the heading as an arrow on a dial and the orange point $(\cos\theta, \sin\theta)$ on the circle, with its value written below. Right, top: the explicit coordinate θ against the heading turned so far, from 330° to 30°. Right, bottom: the implicit numbers $\cos\theta$ (orange) and $\sin\theta$ (green). Frame 'heading 360° = 0°': θ drops from 2π to 0 (a dotted line marked 'θ jumps by 2π'), while $\cos\theta$ and $\sin\theta$ continue smoothly. Last frame, heading 30°: θ has one jump, $\cos\theta$ and $\sin\theta$ have none

Using more numbers than the degrees of freedom, tied by rules, is an **implicit representation** (G-2628) (MR §2.3.2). Its cost is extra numbers; its gain is that it has no jumps and no bad points anywhere. Modern Robotics uses implicit representations throughout for that reason (MR §2.3.2).

An explicit parametrization can be worse than a jump. Latitude and longitude on the surface of a ball misbehave near the North Pole (MR §2.3.2). Walking 1 m east changes the longitude by the angle of a 1 m arc on the circle round the pole, which is arc divided by radius, the arc rule of RO-001:

- ρ is the distance from the pole, in m;
- $\Delta\lambda$ is the change in longitude for 1 m walked, in rad.

$$\Delta\lambda = 1/\rho$$

$$\rho = 10 : \Delta\lambda = 0.1 \text{ rad} = 5.7°$$

$$\rho = 1 : \Delta\lambda = 1 \text{ rad} = 57.3°$$

Figure 12 draws the two walks. Watch the two thick arcs: each stands for 1 m walked (the figure is not to scale), but the orange one, 1 m from the pole, turns the longitude ten times as far as the blue one. The closer to the pole, the faster the longitude changes, without any limit, and at the pole itself every longitude names the same point.

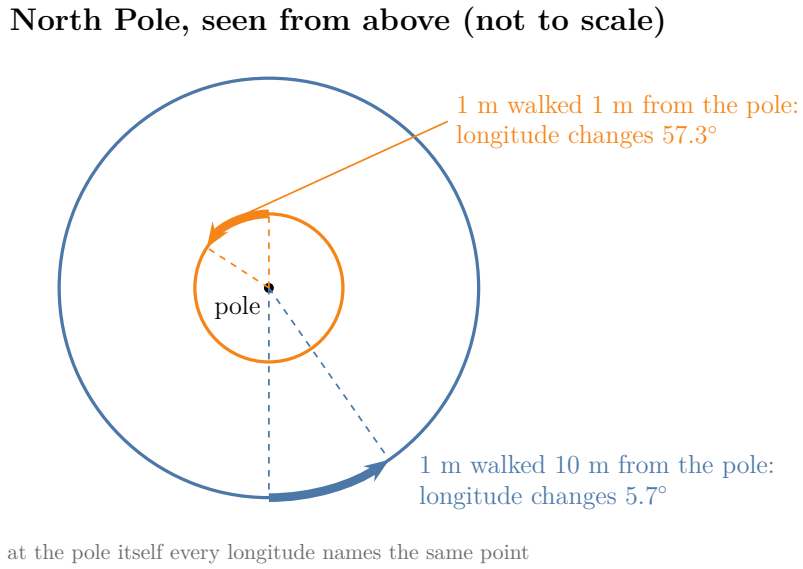


Figure 12: Near the North Pole, seen from above, not to scale. Blue: on the circle 10 m from the pole, 1 m walked east changes the longitude by 5.7 degrees. Orange: on the circle 1 m from the pole, 1 m walked changes it by 57.3 degrees. At the pole itself every longitude names the same point

The sphere looks the same everywhere, at the pole as on the equator, so the trouble is in the numbers, not in the space (MR §2.3.2). The implicit representation, three numbers for the point's position in space with one rule, that its distance from the centre is 1, has no such bad point (MR §2.3.2). Robots meet the sphere as the C-space of a pendulum that can swing in every direction (MR §2.3.1), and planners meet the heading's jump whenever they measure how far apart two configurations are, which needs the distance between two headings measured the short way round.

Every configuration can now be written down as numbers, with or without jumps. What is left is the question the C-space was built for: how to plan a motion through it.

6. The basic motion planning problem

> **Key point:** The third question is the planning problem itself: find a path in the C-space from the start configuration to the goal configuration along which the robot touches no obstacle. The question is the basic motion planning problem, also called the piano mover's problem.

We can now restate the opening problem. The robot's body is a disk with a heading, and every placement of it is one point of the box of Figure 2. The robot starts at the configuration (2, 1, 0.524) and should end in the doorway, with its centre at the doorway's centre, (3.6, 4.0). The grid routes stopped at the centre of the cell under the door, (3.75, 3.75), because a grid route can only stop at a cell centre; a configuration can be any point, so from here on the goal is the doorway's centre.

Figure 13 plays the simplest motion that does it, in the room and in the C-space at once:

1. turn on the spot from 30° to 61.9°, to face the doorway's centre;
2. drive 3.4 m straight to (3.6, 4.0).

The turn and the drive come from how far the doorway's centre lies along x and along y:

- Δx is the run of the drive, in m;
- Δy is its rise, in m;
- atan2 (G-2325) turns the rise and the run into the direction of the drive.

$$\Delta x = 3.6 - 2.0 = 1.6 \text{ m}$$

$$\Delta y = 4.0 - 1.0 = 3.0 \text{ m}$$

$$\text{length} = \sqrt{1.6^2 + 3.0^2} = 3.4 \text{ m}$$

$$\text{heading} = \text{atan2}(3.0,\ 1.6) = 61.9°$$

The robot ends facing along its drive, so the goal configuration is (3.6, 4.0, 1.081), with 61.9° written as 1.081 rad. Watch the right panel of Figure 13. In the frames "turn on the spot", only θ changes, so the point moves straight up. In the frames "drive straight", θ stays the same, so the point moves level. The last frame shows the whole motion as one connected path from the grey start point to the green goal point. Along it, the robot's centre keeps at least 0.382 m from every obstacle and wall, passing closest to the table's corner, so the 0.25 m body clears the table by 0.132 m (notebook; obstacles in C-space shows the same drive in its C-space):

$$0.382 - 0.25 = 0.132 \text{ m}$$

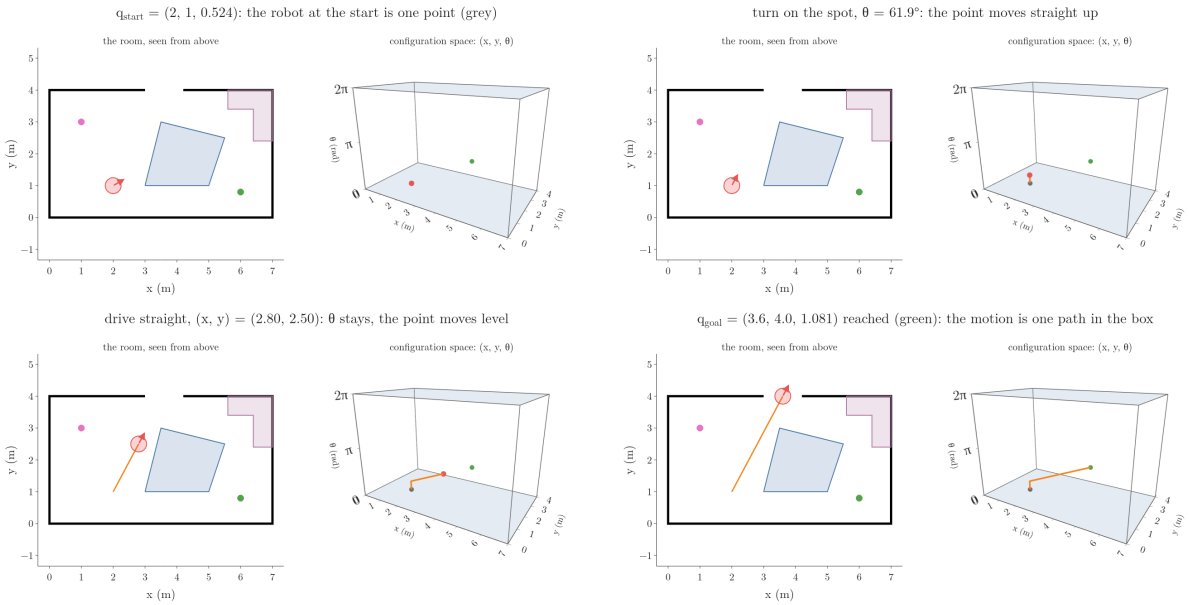


Figure 13: The motion to the doorway. Left: the room seen from above, the robot (red disk with heading arrow) and its orange trail. Right: the configuration space box with the same motion as an orange path. First frame: the robot at q start = (2, 1, 0.524), one grey point. Frames 'turn on the spot': the heading turns from 30 to 61.9 degrees and the point moves straight up. Frames 'drive straight': the robot drives 3.4 m and the point moves level. Last frame: the robot at q goal = (3.6, 4.0, 1.081), the green point, its centre at the doorway's centre; the whole motion is one path in the box

A motion of the robot is a path in its C-space: a continuous curve of configurations, one for each moment, from the start to the goal (LaValle §4.3.1). The robot touches no obstacle exactly when every configuration on the path is one in which the body overlaps nothing; the set of such configurations is the **free space** (G-2651), which the next Note builds. In symbols:

- C is the configuration space, the box of Figure 2 for the floor robot;
- C_{free} is the part of C in which the robot touches no obstacle;
- q_{start} is the start configuration, (2, 1, 0.524);
- q_{goal} is the goal configuration, (3.6, 4.0, 1.081);
- s is how far along the path we are, from 0 at the start to 1 at the end;
- $\tau(s)$ is the configuration at s; the whole curve τ is the path in configuration space;
- $\in$ means "is one of": $\tau(s) \in C_{\text{free}}$ says the configuration at s is free.

The planner must find a path with

$$\tau(0) = q_{\text{start}}$$

$$\tau(1) = q_{\text{goal}}$$

$$\tau(s) \in C_{\text{free}} \text{ for every } s$$

or report correctly that no such path exists (LaValle §4.3.1, Formulation 4.1; MR §10.1). The task of finding such a path is the **basic motion planning problem** (G-2630). The problem is also called the piano mover's problem (MR §10.1; LaValle §4.3.1): moving a piano through a tight doorway is pure geometry, and the piano often has to turn on the way, so its heading is part of the path just as the robot's θ is. Solving it exactly is very hard in general: it is **PSPACE-hard** (G-2559), at least as hard as every problem that a computer can solve with memory growing like a power of the input size, because the number of C-space dimensions grows with the robot (LaValle §4.3.1).

The opening question has its answer: the robot's motion to the door is a path through a three-dimensional C-space, from (2, 1, 0.524) to (3.6, 4.0, 1.081), and the route of the earlier Notes was its shadow on the floor. What is still unknown is which part of the box is free.

7. Summary

Idea	What it is	Why
Configuration q	the numbers that place every point of a robot	(2, 1, 0.524) places the whole floor robot, not just its centre
Configuration space	all configurations; one point per placement	planning becomes finding a path through one space
Degrees of freedom	the fewest numbers that place a robot: freedoms minus independent constraints	3 for the floor robot, 2 for the arm, 5 for both together
Grübler's formula	$m(N-1-J)+\sum f_i$	counts the DOF of linked bodies without counting points
Shape, Cartesian product	what survives bending and stretching; spaces combined pair by pair	the floor robot's C-space is $\mathbb{R}^2 \times S^1$, the arm's a torus $S^1 \times S^1$
Manifold	a space that looks flat near every point	a robot can move a little in each of its directions anywhere
Explicit vs implicit	fewest numbers vs more numbers with rules	the fewest numbers jump at 2π or break at a pole; rules remove that
Basic motion planning problem	a path from start to goal through free configurations	the opening task, stated in C-space

- The robot's motion to the door is a path in its three-dimensional configuration space, from (2, 1, 0.524) to (3.6, 4.0, 1.081), because a placement of the whole body needs its centre and its heading; along the turn and the 3.4 m drive of Figure 13 its body clears the table by 0.132 m.
- The **configuration** (G-2286) of the floor robot is (x, y, θ), because rotating by θ and shifting by (x, y) fixes every other point of the rigid body, such as the front point at (2.2165, 1.125).
- The **configuration space** (G-2287) of the floor robot is a box whose top and bottom faces are the same headings, so the point of the robot driving a full circle leaves through the top face after 5.7596 s and comes back at the bottom.
- The **degrees of freedom** (G-2310) are freedoms minus independent constraints, 3 for the robot's body in the plane (2, 1 and 0 for three of its points), because each fixed distance removes one freedom; the wheels' sideways rule removes none, because it limits velocities, not configurations.
- **Grübler's formula** (G-2620) gives 2 for the arm and 1 for a closed loop of three links, because each revolute joint removes 2 of a link's 3 freedoms; it holds only when the joints' constraints are independent.

- Spaces are **topologically equivalent** (G-2622) when bending and stretching, without cutting or gluing, turns one into the other, so the circle of headings is not the same shape as a line or an interval.
- The **Cartesian product** (G-2625) builds C-spaces from simple ones, so two joint circles make a **torus** (G-2624), not a sphere, and the robot carrying the arm has a five-dimensional C-space.
- Every C-space here is a **manifold** (G-2626), flat near every point, so a robot can move a little in each of its directions from any configuration.
- An **explicit parametrization** (G-2627) uses the fewest numbers and jumps by 2π (-6.24828 rad for a $2°$ turn) or misbehaves at a pole, because a circle, a torus or a sphere cannot be laid flat without cutting; an **implicit representation** (G-2628) such as $(\cos\theta, \sin\theta)$ uses more numbers with a rule and never jumps.
- The **basic motion planning problem** (G-2630), or piano mover's problem, asks for a path τ from the start to the goal through free configurations, because a motion of the whole robot is exactly such a curve.

The robot can now describe every placement of its body as one point of its C-space, and state its motion to the door as a path through that space. Still missing: which configurations are free. The table and the cabinet are shapes in the room, not in the C-space; obstacles in C-space grows them by the robot's 0.25 m body to find the free space.

8. Sources

Built from

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- LaValle, S. M. (2006). *Planning Algorithms*. Cambridge University Press. §4.1.2 "Manifolds" and "Cartesian products", §4.3.1 "Definition of the Basic Motion Planning Problem" (Formulation 4.1, the piano mover's problem). Free online: https://lavalle.pl/planning/node131.html (LaValle)

Other references

- LaValle, S. M., "Motion Planning: Life in C-Space", ICRA invited tutorial, YouTube, https://www.youtube.com/watch?v=ERaZJru80bM

9. Key terms

Terms taught in this Note come first; linked terms are recaps, taught in the Note the link opens.

Term	Meaning
Configuration (G-2286)	The list of numbers that fixes the position of every point of a robot, such as $q = (x, y, \theta) = (2, 1, 0.524)$ for a floor robot or the joint angles (θ_1, θ_2) of a 2R arm; with it a planner treats the whole robot as one point.
Configuration space (C-space) (G-2287)	The set of all configurations of a robot, one point per placement of the whole robot, such as a box of (x, y, θ) whose top and bottom faces are the same headings for a floor robot; motion planning looks for paths through it.
Degrees of freedom (of a robot) (G-2310)	The smallest number of real numbers that places a robot, which is also the number of dimensions of its C-space: the freedoms of its points minus the independent constraints on them, such as 3 for a floor robot (x, y, θ) and 2 for a 2R arm; it tells a planner how many numbers it must search over.
Grübler's formula (G-2620)	The count of a mechanism's degrees of freedom from its links and joints, $m(N - 1 - J) + \sum f_i$ ($m = 3$ in the plane, N links with the ground, J joints, f_i freedoms of joint i), such as 2 for a 2R arm; it holds when the joints' constraints are independent.
Topologically equivalent (G-2622)	Said of two spaces when one can be deformed into the other by bending and stretching, without cutting or gluing, such as a circle and an ellipse, or a coffee mug and a doughnut; equivalent spaces have the same shape for planning.

Term	Meaning
Cartesian product (of spaces) (G-2625)	The space $X \times Y$ of every pair made of one point of X and one point of Y, with as many dimensions as both together, such as $\mathbb{R}^2 \times S^1$ for a floor robot's position and heading; it builds a robot's C-space from the spaces of its parts.
Torus T^2 (G-2624)	The surface of a doughnut, the Cartesian product of two circles $S^1 \times S^1$; it is the C-space of a 2R arm, one point per pair of joint angles.
Manifold (G-2626)	A space in which a small neighbourhood of every point looks like a piece of flat space of the same dimension (a line, a plane), such as a circle or a torus but not a T-junction; near any configuration a robot can then move a little in each of its directions.
Explicit parametrization (G-2627)	Writing an n-dimensional space with exactly n numbers, such as a heading θ from 0 to 2π or latitude and longitude on a sphere; it uses the fewest numbers but jumps or breaks somewhere on a space that is not flat.
Implicit representation (G-2628)	Writing a space with more numbers than its dimension, tied by rules, such as a heading as $(\cos\theta, \sin\theta)$ with $\cos^2\theta + \sin^2\theta = 1$; it costs extra numbers but has no jumps and no bad points such as a pole.
Basic motion planning problem (piano mover's problem) (G-2630)	Given a robot, its obstacles, a start configuration and a goal configuration, find a continuous path in C-space between them through configurations where the robot touches no obstacle, or report that none exists; it states every collision-free motion task as one search through C-space.
Vector (G-2081)	A list of numbers: a 1D tensor.
Pose (G-2280)	Where a robot is and which way it faces: on a floor, its position (x, y) and heading θ, such as (2 m, 1 m, 30°); needed because a robot at one spot drives off differently depending on its heading.

Term	Meaning
Angle wrapping (G-2404)	Adding or subtracting full turns until an angle or angle difference lies between $-180°$ and $180°$, so that $175°$ and $-178°$ count as $7°$ apart, not $353°$; needed whenever bearings or headings are compared.
atan2 (G-2325)	A function atan2(rise, run) that returns the direction of a line over the full circle from its rise and run given separately; needed because the plain inverse tangent of the ratio cannot tell opposite directions apart.
Free space (G-2651)	All configurations at which the robot touches no obstacle, C_{free}, the C-space without the obstacle region; a planner looks for a path that stays inside it.
PSPACE-hard (G-2559)	Said of a problem at least as hard as every problem solvable with memory growing like a power of the input size (PSPACE, which contains NP); the general motion planning problem and Sokoban are PSPACE-hard, which is why practical planners give up exactness.